

APPENDIX
SUPPLEMENTARY MATERIALS

Supplementary Materials consist of the following sections:

- Policy implementation and training objective in Sec. A.
- Prediction metric definitions and supplementary stiffness results in Sec. B.
- Grasping-force signals and evaluation metrics in Sec. C.
- VLM prompt and illustrative response in Sec. D.

A. Policy Implementation and Training Objective

Index Embeddings. For the property token in Sec. IV-F2, predictor labels use indices 0–2, while policy embeddings shift mass and friction to 1–3 and reserve 0 for unavailable pre-contact properties. The contact indicator and discretized properties are embedded together. This mapping distinguishes the lowest property class from an unavailable observation.

Training Objective. The ACT-style policy uses the weighted objective

$$\mathcal{L}_{\text{policy}} = w_1 \mathcal{L}_{\text{action}} + w_2 \mathcal{L}_{\text{KL}}, \quad (8)$$

where w_1, w_2 are loss weights. The action loss $\mathcal{L}_{\text{action}}$ is the ℓ_1 error between predicted and demonstrated action sequences. The Kullback–Leibler loss $\mathcal{L}_{\text{KL}}$ regularizes the latent action-style distribution toward a Gaussian prior. This latent regularization is separate from the discrete property-index replacements described in Sec. IV-F2.

B. Prediction Metrics and Supplementary Results

For a classification task, let $\hat{y}_{et}$ and y_{et} be the predicted and reference labels of window t in episode e . With E test episodes and T_e windows in episode e , the episode-balanced accuracy reported in the main paper is

$$A_{\text{ep}} = \frac{1}{E} \sum_{e=1}^E \frac{1}{T_e} \sum_{t=1}^{T_e} \mathbf{1}[\hat{y}_{et} = y_{et}]. \quad (9)$$

Here $\mathbf{1}[\cdot]$ is the indicator function. Each episode has equal total weight. For class c , let TP_c , FP_c , and FN_c denote true positives, false positives, and false negatives. Macro-F1 averages the class-specific F1 scores over $C = 3$ classes:

$$F_{1,\text{macro}} = \frac{1}{C} \sum_{c=1}^C \frac{2\text{TP}_c}{2\text{TP}_c + \text{FP}_c + \text{FN}_c}. \quad (10)$$

For N evaluated windows with positive stiffness targets s_i and predictions $\hat{s}_i$, the regression metrics are

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{s}_i - s_i|, \quad (11)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{s}_i - s_i)^2}, \quad (12)$$

$$\text{MAPE} = \frac{100}{N} \sum_{i=1}^N \frac{|\hat{s}_i - s_i|}{|s_i|}. \quad (13)$$

MAE and RMSE use N/mm; MAPE is a percentage. Pearson correlation uses Eq. (20) with measured and predicted stiffness

TABLE VII
Supplementary stiffness metrics from the unabridged results. Coverage W_{10} and W_{20} is in percent.

Setting	Pearson r	W_{10}	W_{20}
<i>Evaluation protocols</i>			
Seen-object	0.980	84.22	96.56
Held-out-object	0.947	61.56	89.64
One-shot	0.680	49.47	75.19
<i>Encoder-branch and text ablations</i>			
w/o tactile content	0.769	46.39	80.69
w/o visual content	0.887	45.97	79.79
w/o visual flow	0.942	59.70	88.39
w/o tactile flow	0.950	70.87	89.52
w/o both flow	0.633	56.94	73.47
Text only	0.580	42.37	70.00
w/o text	0.900	56.46	77.36
<i>Loss ablations</i>			
Manual loss weights	0.922	59.51	83.06
DWA loss weights	0.880	54.72	84.44
Cross entropy	0.945	59.38	85.14
Full model (held-out-object)	0.947	61.56	89.64

in place of human and robot signals. For relative tolerance ϵ , the coverage is

$$W_{\epsilon} = \frac{100}{N} \sum_{i=1}^N \mathbf{1} \left[\frac{|\hat{s}_i - s_i|}{|s_i|} < \epsilon \right]. \quad (14)$$

The columns W_{10} and W_{20} below use $\epsilon = 0.10$ and 0.20 . Table VII retains the additional stiffness metrics from the unabridged manuscript for the protocols and ablations in Tables II–IV. The values are transcribed from that version, without recomputing predictions.

Removing tactile flow increases W_{10} from 61.56% to 70.87%, while mass and friction classification worsen (Table III). The supplementary results thus reinforce the task-dependent behavior of the input branches.

C. Grasping-Force Metrics

1) *Signals and comparison set:* Using the fingertip calibration in Sec. III, the normalized grasp signal and calibrated force are

$$g = \sum_q S_q, \quad F_{\text{grasp}} = \sum_q (a_q S_q + b_q), \quad (15)$$

where q ranges over the thumb, index, and middle fingers. Table VI uses g ; Fig. 7 uses F_{grasp} in newtons.

Let x_{mit} be the mean normalized signal over the stable final 3 s of clean-success trial t for method m on object i , and h_i the corresponding mean human teleoperation reference. Let $\mathcal{T}_{mi}$ contain the retained trials. For methods a, b and object split D , the common-success set is

$$\mathcal{C}_{ab,D} = \{i \in D : |\mathcal{T}_{ai}| > 0, |\mathcal{T}_{bi}| > 0\}. \quad (16)$$

Write $K = |\mathcal{C}_{ab,D}|$. The evaluated-object ratio is $K/|D|$, with $|D| = 40$ for ID and 12 for OOD. It measures coverage of the force comparison, not trial success rate or Pairwise agreement.

2) *Absolute error and object averages:* The trial-level absolute error and object-level robot mean are

$$N_m = \sum_{i \in \mathcal{C}_{ab,D}} |\mathcal{T}_{mi}|, \quad (17)$$

$$E_m = \frac{1}{N_m} \sum_{i \in \mathcal{C}_{ab,D}} \sum_{t \in \mathcal{T}_{mi}} |x_{mit} - h_i|, \quad (18)$$

$$r_{mi} = \frac{1}{|\mathcal{T}_{mi}|} \sum_{t \in \mathcal{T}_{mi}} x_{mit}. \quad (19)$$

Absolute errors are taken before averaging trials. Different methods may retain different numbers of trials on the same object. Correlation and ordering instead give each common object one robot mean.

3) *Pearson correlation:* With $\bar{h} = K^{-1} \sum_i h_i$ and $\bar{r}_m = K^{-1} \sum_i r_{mi}$, Pearson correlation is

$$\rho_m = \frac{\sum_i (h_i - \bar{h})(r_{mi} - \bar{r}_m)}{\sqrt{\sum_i (h_i - \bar{h})^2 \sum_i (r_{mi} - \bar{r}_m)^2}}, \quad (20)$$

where all sums range over $\mathcal{C}_{ab,D}$. It measures linear association and is undefined if either vector has zero variance. High correlation need not imply small absolute error.

4) *Pairwise ordering:* Eligible unordered object pairs exclude ties in either signal:

$$\mathcal{P}_m = \{(i, j) : i, j \in \mathcal{C}_{ab,D}, i < j, h_i \neq h_j, r_{mi} \neq r_{mj}\}. \quad (21)$$

Their ordering agreement is

$$Q_m = \frac{1}{|\mathcal{P}_m|} \sum_{(i,j) \in \mathcal{P}_m} \mathbf{1}[(h_i - h_j)(r_{mi} - r_{mj}) > 0]. \quad (22)$$

A pair agrees when the robot and human apply greater force to the same object. The score lies in $[0, 1]$ and is undefined if no eligible pair remains. Robot ties can produce different eligible-pair counts for methods evaluated on the same objects.

5) *Force curves grouped by GT properties:* Figure 7 uses each method’s own clean-success trials. If $\mathcal{B}_{mpb}$ contains its trials for property p and GT bin b , the bin mean is

$$\mu_{mpb} = \frac{1}{|\mathcal{B}_{mpb}|} \sum_{(i,t) \in \mathcal{B}_{mpb}} z_{mit}, \quad (23)$$

where z_{mit} is the stable-interval mean calibrated force. These sets are not restricted to $\mathcal{C}_{ab,D}$. GT bins organize the analysis; ViTacPhys still supplies predicted properties during execution. The human curve is a reference behavior, not a force target uniquely determined by the property labels.

D. VLM Prompt and Illustrative Response

The boxes preserve the supplied prompt and rubber-duck response, with paragraph reflow and highlighted headings. Confidence values are generated hypotheses, not calibrated probabilities or measured labels. The full-container assumption is dataset context supplied to the VLM.

Input-context discrepancy. The supplied prompt includes post-grasp video and held-shape instructions, whereas Sec. IV-C

specifies one pre-grasp image. This example documents the supplied text; it does not establish which prompt was used for the reported single-image experiments. The original wording is retained so the discrepancy remains explicit.

Prompt: observation and reasoning instructions

The image shows a object will be grasped.

Important observation rule:

Focus ONLY on the object being held by the gripper or hand. Ignore the robot hand, gripper, sensors, cameras, or any other data collection equipment. Ignore background objects such as walls, tables, or devices. Only analyze the object that is being grasped.

Video context:

The video starts AFTER the object has already been stably grasped. Therefore, you should NOT expect new deformation to occur during the video. If the object already appears flattened, bent, or compressed, that indicates lower stiffness. If the object maintains its shape while being firmly held, that indicates higher stiffness. Use the CURRENT visible shape to infer stiffness.

Dataset context for mass reasoning:

The objects in this dataset are simple everyday containers such as paper cups, plastic bottle, cans, bottles, mug, plush toys or rubber toys. They do NOT contain arbitrary hidden objects. Mass differences mainly come from two sources: 1) liquid inside containers (for example water in a bottle or can); 2) the material itself having higher density (for example glass or metal).

Differences in stiffness and friction mainly come from the material type. For example, a plastic cup and a paper cup can look visually similar, but their stiffness and friction properties are different. When the visual appearance is ambiguous, include multiple material hypotheses.

Important rule:

If a container contains liquid, it is usually FULL in this dataset. A liquid-filled container should therefore be considered HIGH mass and increased stiffness.

If the material itself is dense (glass or metal), estimate the mass using both material and object size. Do NOT assume unknown hidden objects inside.

Write TWO parts.

Part 1: Understanding (2 to 4 short sentences). Describe what the held object likely is and what visible physical cues support this understanding. Mention cues such as thin wall, rigid outline, compression marks, glossy surface, slip signs, or surface texture. If uncertain, mention up to two plausible object types. Do NOT describe the robot hand, sensors, or background objects.

Part 2: Physical reasoning summary for downstream encoding. Follow the exact format below. Use natural language sentences and stable structure. Do NOT output JSON, bullet lists, or additional commentary.

Each hypothesis MUST follow this exact pattern:

<label> : confidence <0.X> ; evidence <short visual cues>

Use two hypotheses for each property whenever possible. Confidence values must be decimals between 0.0 and 1.0. Evidence should contain short visible cues only.

Allowed labels:

Mass: low, medium, high
Materials: paper, thin plastic, rigid plastic, rubber, foam, fabric, plush, glass, thin metal, rigid metal
Stiffness: low, medium, high
Friction: low, medium, high

Reasoning rules:

- Do NOT output a single definitive conclusion.
- Provide multiple plausible hypotheses when uncertainty exists.
- If the object visually appears light but could contain liquid, include a HIGH mass hypothesis.
- If the object visually resembles paper but could also be plastic, include both materials.
- Infer stiffness from the current shape while being held.

Prompt: observation and reasoning instructions (continued)

- Mass reasoning should rely mainly on visible size, wall thickness, material cues, and the possibility of liquid inside.

Prompt: required output structure**Physical reasoning summary:**

Object type: <short noun phrase describing the held object>.

Mass hypotheses:

<mass_label> : confidence <0.X> ; evidence <visual cues>;

<mass_label> : confidence <0.X> ; evidence <visual cues>.

Material hypotheses:

<material> : confidence <0.X> ; evidence <visual cues>;

<material> : confidence <0.X> ; evidence <visual cues>.

Stiffness hypotheses:

<stiffness_level> : confidence <0.X> ; evidence <visual cues>;

<stiffness_level> : confidence <0.X> ; evidence <visual cues>.

Friction hypotheses:

<friction_level> : confidence <0.X> ; evidence <visual cues>;

<friction_level> : confidence <0.X> ; evidence <visual cues>.

Prompt: in-context example**Physical reasoning summary:**

Object type: disposable cup.

Mass hypotheses:

low : confidence 0.6 ; evidence small size thin wall;

high : confidence 0.4 ; evidence possible liquid inside container.

Material hypotheses:

paper : confidence 0.5 ; evidence matte thin wall;

thin plastic : confidence 0.5 ; evidence glossy smooth surface.

Stiffness hypotheses:

low : confidence 0.6 ; evidence visible compression while held;

medium : confidence 0.4 ; evidence shape still partially maintained.

Friction hypotheses:

medium : confidence 0.7 ; evidence paper or plastic surface;

low : confidence 0.3 ; evidence smooth glossy surface.

Supplied response: rubber duck

Part 1: Understanding The held object appears to be a small yellow rubber duck toy with an orange beak and printed black eyes. Its surface looks smooth and slightly matte, and the body keeps a rounded shape without visible denting while resting in the hand, suggesting a hollow but shape-retaining squeeze toy. The size relative to the hand suggests it is lightweight and not a liquid-filled container.

Part 2: Physical reasoning summary:

Physical reasoning summary:

Object type: rubber duck squeeze toy.

Mass hypotheses:

low : confidence 0.85 ; evidence small toy size hollow-looking rounded body;

medium : confidence 0.15 ; evidence could be thicker rubber body than typical.

Material hypotheses:

rubber : confidence 0.8 ; evidence smooth molded surface classic squeeze-toy appearance;

rigid plastic : confidence 0.2 ; evidence uniform shape and clean molded seams could be hard plastic.

Stiffness hypotheses:

medium : confidence 0.65 ; evidence maintains rounded shape with no visible collapse;

low : confidence 0.35 ; evidence typical duck toys are squeezable hollow rubber.

Friction hypotheses:

medium : confidence 0.7 ; evidence rubbery matte surface likely provides grip;

low : confidence 0.3 ; evidence smooth surface could slide if plastic-like.